

Aliah University

End Semester Examination (Spring Semester) 2025

(For 2nd Year 4th Semester B.Tech(CSE))

Paper Name: Computer Organization and Architecture

Paper Code: CSEUGPC06

Full Marks: 80

Time: 3 hours

Group A(Answer all questions)

5 X2=10

1. A two byte relative mode branch instruction is stored in memory location 1000. The branch is made to the location 87. What is the effective address? [CO2, Apply]
2. What are the functions of PC and MAR? [CO1, Remember]
3. What is locality of reference? [CO3, Remember]
4. What is biased exponent? [CO2, Remember]
5. Define Instruction cycle and Machine cycle. [CO1, Remember]

Group B(Answer any 5 questions)

(5X6=30)

1. Describe the different types of instructions with examples. [CO2, Understand]
2. Explain with diagram the IAS Computer. [CO1, Understand]
3. Differentiate i) Hardwired Control and Microprogrammed Control. (3+3) [CO3, Understand]
ii) SRAM vs DRAM.
4. Represent the decimal value -7.5 in IEEE -754 single precision floating-point format. [CO2, Apply]
5. How many 256X4 RAM chips are needed to provide a memory capacity of 2048 bytes? Show also the corresponding interconnection diagram. (3+3) [CO3, Apply]
6. a) Explain how pipelining improves performance in computer architecture? [CO5, Understand]
b) A non-pipelined processor takes 5 cycles to execute an instruction. If pipelining is introduced with 5 stages and ideal conditions, calculate the speedup and efficiency for 100 instructions. (3+3) [CO5, Apply]

Group C(Answer any 4 questions)

(4X10=40)

1. a) Describe Booth's Multiplication Algorithm and explain it. [CO1, Understand]
b) Multiply 7 and 3 using Booth's Multiplication upto four digits. [CO1, Apply] (5+5)

2. a) What is Instruction format?

[CO1,Remember]

b) Draw and Explain the state transition diagram of an instruction cycle. [CO1,Understand]

c) Discuss the following addressing mode with examples: i) Immediate ii) Register [CO2,Understand]

(2+4+4)

3. Evaluate the arithmetic statement $X=(A+B)/(C-D)$ using zero, one, two and three address instructions. [CO2,Evaluate]

(2.5X4)

4. a) A computer has 512 KB cache memory and 2 MB main memory. If the block size is 64 bytes, then find out the subfields for

i) direct mapped cache

ii) associative

iii) 8-way set associative cache.

[CO3,Apply]

b) A hierarchical cache-main memory subsystem has the following specifications (i) Cache access time of 160 ns (ii) main memory access time of 960 ns (iii) hit ratio of cache memory is 0.9. Calculate the following: (a) Average access time of the memory system. (b) Efficiency of the memory system.

[CO3,Evaluate] (6+4)

5. a) Explain the various modes of data transfer between the CPU and I/O devices. Compare them with suitable examples. [CO4,Understand]

b) Explain with a neat sketch the working principle of DMA. Write its advantages.

[CO4,Understand] (6+4)

B.Tech Examination 2024-25
Department of Electronics and Communication Engineering
(Even Semester Regular & Supplementary)
Course Title: Principle Of Communication System, Course Code: ECEUGOE02

Full Marks: 80**Time: 3.00 Hrs**

- Answer Question No. 1 and any 7 Questions from Question No. 2 to 12
- Figures on the right hand side margin indicate full marks.
- Answer all parts of a question in same place.
- Symbols have their usual meaning.

1. Answer any five questions.

- | | |
|---|---|
| (a) What is baseband transmission? | 2 |
| (b) What is modulation? | 2 |
| (c) Define transmission efficiency of an AM system. | 2 |
| (d) Define frequency deviation in FM. | 2 |
| (e) Write the properties of line coding. | 2 |
| (f) What is a lossless channel? | 2 |
| (g) State Shannon-Hartley Law of a communication channel. | 2 |

- | | |
|---|---|
| 2. (a) Draw the block diagram of a communication system and briefly explain the function of each block. | 5 |
| (b) Mention different types of modulation techniques. | 2 |
| (c) Why do we need modulation in communication? | 3 |

- | | |
|--|---|
| 3. (a) What is meant by the term 'amplitude modulation'? | 2 |
| (b) Define modulation index for AM wave. | 2 |
| (c) Derive an expression for single-tone amplitude modulated wave. | 4 |
| (d) Write the condition for over modulated AM wave and draw the waveform of an over modulated AM wave. | 2 |

- | | |
|---|---|
| 4. (a) Prove that the balanced modulator produces an output signal consisting of two side band only with the carrier removed. | 5 |
| (b) Derive an expression of transmission efficiency of a single tone amplitude modulated signal. | 3 |
| (c) Calculate the percentage of power savings for DSB-SC signal for the percentage modulation of 50%. | 2 |

- | | |
|--|---|
| 5. (a) What is frequency modulation? | 2 |
| (b) Derive an expression for single tone frequency modulated signal. | 4 |
| (c) Explain the generation of narrowband FM signals with suitable block diagram. | 4 |

- | | |
|--|---|
| 6. (a) What is Carson's rule? | 2 |
| (b) Explain the operation of balanced slope detector. | 5 |
| (c) A single tone frequency modulated wave is denoted by following expression $s(t) = 2 \cos[2\pi \times 10^6 t + 10 \sin(2\pi \times 10^3 t)]$. Determine carrier frequency, modulating frequency and maximum frequency deviation. | 3 |

- | | |
|--|---|
| 7. (a) State and Prove sampling theorem for a band limited signal. | 6 |
| (b) What is aliasing effect? How is it removed? | 2 |
| (c) Determine the Nyquist rate for a continuous time signal $x(t) = 6\cos 50\pi t + 20\sin 300\pi t - 10\cos 100\pi t$ | 2 |

8. (a) How is an analog signal converted to a digital signal in PCM system? 5
 (b) Derive the expression for transmission bandwidth in a PCM system. 3
 (c) A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that number of quantization level is 512. Determine transmission bandwidth and final bit rate. 2
9. (a) Explain companding in non-uniform quantization? 4
 (b) Draw and explain compressor characteristics. 3
 (c) Describe about μ -law companding. 3
10. (a) Explain the operation of a delta modulator with suitable block diagram. 5
 (b) What is slope overload distortion and how is it avoided in ADM? 2
 (c) Given a sine wave of frequency f_m and amplitude A_m applied to a delta modulator having a step size Δ . Show that slope overload distortion will occur if $A_m > \frac{\Delta}{2\pi f_m T_s}$, Here T_s is the sampling frequency 3
11. (a) Consider the binary data sequence 1011101011. Draw the waveform for the following signalling format: (i) Unipolar RZ (ii) Split Phase Manchester coding (iii) AMI 3
 (b) Sketch the waveform of ASK, FSK and PSK signal for the input binary sequence 1101010010. 3
 (c) Explain briefly the coherent detection of BFSK signal with suitable block diagram. 4
12. (a) Define entropy, information rate and how are they related? 3
 (b) A discrete memory less source (DMS) X has five possible outcome with probabilities $P_1 = \frac{1}{2}$, $P_2 = \frac{1}{4}$, $P_3 = \frac{1}{8}$, $P_4 = \frac{1}{16}$, $P_5 = \frac{1}{16}$. Find entropy and information rate if there are 16 outcomes per second. 4
 (c) A DMS X has five symbols x_1, x_2, x_3, x_4 and x_5 with $P(x_1) = 0.4$, $P(x_2) = 0.19$, $P(x_3) = 0.16$, $P(x_4) = 0.15$ and $P(x_5) = 0.1$. Construct a Huffman Code for the source. 3

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End-Semester Examination - 2025 (Spring)
MATUGBS05
(Probability and Statistics)
Semester – IV (4-year B.Tech in Computer Science and Engineering (CSE))
Full Marks – 80 **Time – 3:00 Hours**

Notations and symbols have their usual meaning.

Section A. Answer ALL questions. Each question carries Two marks.

1. Two unbiased coins are tossed. What is the probability of getting at most one head?
 (a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{6}$ (d) $\frac{1}{3}$
2. Company A produces 10% defective products, Company B produces 20% defective products, and C produces 5% defective products. If choosing a company is an equally likely event, then find the probability that the product chosen is defective.
 (a) 0.11 (b) 0.21 (c) 0.22 (d) 0.12
3. If the S.D. of a set of observations is 4, and if each observation is divided by 4, find the S.D. of the new observations.
 (a) 4 (b) 3 (c) 2 (d) 1
4. The probability that at least one of the events Q and R occur is 0.6. If Q and R have probability of occurring together as 0.2, then $P(\bar{Q}) + P(\bar{R})$ is?
 (a) 1.2 (b) 0.8 (c) 0.4 (d) Indeterminate
5. If a random variable X satisfies the Poisson distribution with mean value of 3, then the probability that the $X \geq 2$,
 (a) $1 - 2e^{-3}$ (b) $1 + 4e^{-3}$ (c) $4e^{-3}$ (d) $1 - 4e^{-3}$

Section B. Answer ALL questions. Each question carries Two marks.

- ✓ 6. Define Type-I and Type-II errors.
7. Explain Simple and composite hypothesis.
- ✓ 8. Explain Level of significance and power function of a test.
9. Define normal probability distribution.
- ✓ 10. If T is an unbiased estimator for θ , show that T^2 is a biased estimator for θ^2 .

Section C. Answer any Six questions. Each question carries Ten marks.

- ✓ 11. (a) A box contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box at random. Find the probability that among the balls drawn, there is at least one ball of each colour.

- (b) If A, B and C are pairwise independent events and A is independent of $(B \cup C)$, then show that A, B and C are mutually independent. [5+5]

12. (a) If X and Y are two random variables, then show that $E(X + Y) = E(X) + E(Y)$, provided all the expectations exist.
 (b) Show that the moment generating function of the sum of a number of independent random variables is equal to the product of their respective moment generating functions. [5+5]

✓ 13. A random variable X has the following probability distribution:

x:	0	1	2	3	4	5	6	7
p(x):	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2 + k$

- (i) Find k, (ii) Evaluate $P(X < 6)$, $P(X \geq 6)$ and $P(0 < X < 5)$, (iii) If $P(X \leq a) > \frac{1}{2}$, find the minimum value of a, and (iv) Determine the distribution function of X. [2+3+2+3]

14. (a) Define the binomial distribution with parameters p and n. Give a situation in real life where the distribution is likely to be realized. Obtain the moment generating function of the binomial distribution.

- ✗ (b) If the independent r.v. X and Y are binomially distributed, respectively with $n = 3$, $p = 1/3$ and $n = 5$, $p = 1/3$. Find $P(X + Y \geq 1)$. [5+5]

15. (a) Find the moment generating function of Poisson distribution.

- (b) State and prove the reproductive property of the Poisson distribution. [5+5]

✓ 16. Two random variables X and Y have the following joint probability density function :

$$f(x, y) = \begin{cases} 2 - x - y & \text{if } 0 \leq x \leq 1; 0 \leq y \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find marginal probability density function of X and Y.

- (b) Conditional density functions.

- (c) $\text{Var}(X)$ and $\text{Var}(Y)$.

- (d) Covariance between X and Y. [10]

17. (a) If X_1 and X_2 are two independent normal variates with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 respectively, show that the variable U and V, where $U = X_1 + X_2$ and $V = X_1 - X_2$, are independent normal variates. Find the mean and variance of U and V.

- (b) In a normal distribution 31% of the items are under 45 and 8% of the items are over 64. Find mean and variance of the distribution. [5+5]

- ✓ 18. (a) Show that correlation coefficient is independent of change of origin and scale.

- (b) Given $n = 10$, $\sigma_x = 5.4$, $\sigma_y = 6.2$ and the sum of the product of the deviation from the mean of X and Y is 66. Find the correlation coefficient. [5+5]

- ✓ 19. (a) The equations of two regression lines are $7x - 16y + 9 = 0$ and $5y - 4x - 3 = 0$. Find the coefficient of correlation and the means of x and y.

- (b) If $\sigma_x = \sigma_y = \sigma$ and the angle between the regression lines is $\tan^{-1}(\frac{4}{3})$. Find r. [5+5]

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BIOUGBS01 Biology for Engineers

2025

Marks-80

Time-3Hrs

1×25=25

A. Answer the following question by selecting the correct options.

1. Anticodon is present in

- (a) DNA (b) tRNA (c) rRNA (d) mRNA

2. Where is DNA present in the eukaryotic cells?

- A. Inside the nucleus B. With other cellular contents C. Inside the ribosomes D. Not present

3. A single strand of mRNA attached to complex of ribosomes is called

- (a) Okazaki fragments (b) polymer (c) polysome (d) polypeptide

4. The EMP pathway in eukaryotes usually takes place in

- (a) nucleus (b) lysosome (c) Golgi apparatus (d) cytosol

5. Which of the following enzyme catalyzes the first step of glycolysis?

- a) Hexokinase b) Pyruvate kinase c) Glucokinase d) Phosphofructokinase-1

6. Book lungs are respiratory organs in

- A. Insects B. Arachnids C. Molluscs D. Echinoderms

7. The crossing of F1 to either of the parents is known as

- A. Test cross B. Back cross C. F1 cross D. All of the above

8. Which is the largest phylum of Animal on the earth.

- A. Mollusca B. Amphibia C. Arthropoda D. Aves

9. *Taenia Solium* is under the Phylum of

- A. Aschelminthes B. Annelids C. Platyhelminths D. Mollusca

10. Which of the following is a flightless bird?

- A. Pigeon B. vulture C. Parrot D. ostrich

11. Who coined the term mitochondria?

- A. Kolliker B. Benda C. Fleenming D. Robert Brown

12. Plasma membrane is

- A. Permeable B. Selectively permeable C. Impermeable D. Semi-permeable

13. The excretory organ in the cockroach is

- (A) green gland (B) malpighian tubules (C) nephridia (D) kidney

14. The mammary are found in

- (A) Aves (B) Mammalia (C) Amphibian (D) Reptile

15. Air bladder is absent in

- (A) Dog fish (B) Catla (C) Pihu (D) Flying fish

16. The plasma membrane is made up with

- A. Lipid B. Protein C. Lipid & Protein D. Protein & Carbohydrate

17. Which of the following immunity is present from our birth?

- A. Innate Immunity B. Active immunity C. Passive immunity D. Acquired immunity

18. Which of the following purine bases is present in RNA?

- (a) Uracil (b) Thymine (c) Cytosine (d) Guanine

19. Which of the following RNAs are the most abundant in an animal cell?

- (a) mRNA (b) tRNA (c) miRNA (d) rRNA

20. In DNA, the enzyme which breaks the H₂ bonds is

- (a) Ligase (b) Helicase (c) Topoisomerase (d) Polymerase

21. Which substrate is used in the last step of glycolysis?

- a) Glyceraldehyde 3-phosphate b) Pyruvate
c) Phosphoenolpyruvate d) 1, 3-bisphosphoglycerate

22. Ctenophores, commonly known as

- (A) Flat worms (B) Sea walnuts (C) Round worms (D) Sponges

23. The canal system is a characteristic feature of

- (A) Arthropods (B) Mollusca (C) Sponges (D) Echinoderms

24. Which of the following is the function of lysosomes?

- A. Autophagy B. Autolysis C. Digestion D. All of the above

25. Which is a cold-blooded animal

- (A) Horse (B) Sea-horse (C) Bat (D) Crane

2×5= 10

B. Answer any 5 questions

- ✓ 1. What is Nephridia? Where is it found?
- ✓ 2. What is Humoral immunity?
3. What is Nucleosome?
- ✓ 4. Mention the location and function of Book Lung.
- ✓ 5. Define Allele.
- ✓ 6. What is Osculum? Mention its one function.
7. Define Codon.

5×5= 25

C. Answer any 5 questions

- ✓ 1. Mention the four differences between antibody and antigen.
2. Describe the steps of the Glycolysis process.
- ✓ 3. Mention the characteristic features of class Amphibia with a suitable example.
- ✓ 4. Write down a short note on the Central Dogma.
- ✓ 5. Mention the differences between Prokaryotic and Eukaryotic cells.
- ✓ 6. Mention the structure of DNA with a suitable diagram.

10×2= 20

D. Answer the following questions

1. Describe the Krebs cycle with a suitable diagram.
2. Describe the structure and function of mitochondria.

**UCCUGMC02 Environmental Science
2025**

Full Marks: 80

Time: 3 Hours

Group-A

(Answer any 20 questions from the following)

(Each question carries 1 mark)

20 × 1 = 20)

1. A consumer that eats only plants is called:
A) Carnivore B) Herbivore C) Omnivore D) Decomposer
2. Solar energy is:
A) Non-renewable B) Renewable C) Finite D) Limited
3. Which is a secondary pollutant?
A) CO₂ B) NO₂ C) PAN (Peroxyacetyl nitrate) D) SO₂
4. Smog is a mixture of:
A) Dust and rain B) Smoke and water vapor C) Smoke and fog D) Dust and oxygen
5. World Biodiversity Day is celebrated on:
A) June 5 B) April 22 C) May 22 D) October 16
6. Which of the following practices leads to land degradation?
A) Terrace farming B) Afforestation C) Overgrazing D) Drip irrigation
7. The most abundant gas in the atmosphere is:
A) Oxygen B) Nitrogen C) Carbon dioxide D) Helium
8. Natural pollutants are:
A) CO₂ from industries B) Volcanic ash C) Plastic D) Fertilizer runoff
9. The zone of atmosphere closest to Earth is:
A) Troposphere B) Stratosphere C) Mesosphere D) Thermosphere
10. Greenhouse effect is due to:
A) Reflection of solar radiation B) Absorption of infrared radiation
C) Absorption of UV rays D) None
11. Which one is a point source of water pollution?
A) Agricultural runoff B) Factory outlet C) Urban runoff D) Storm drains
12. Primary consumers are:
A) Herbivores B) Carnivores C) Omnivores D) Decomposers
13. Food chain starts with:
A) Herbivores B) Carnivores C) Producers D) Decomposers
14. Which of the following has the longest food chain?
A) Grassland ecosystem B) Forest ecosystem C) Desert ecosystem D) Pond ecosystem
15. Energy flow in ecosystem is:
A) Unidirectional B) Circular C) Multidirectional D) Reversible
16. Which one is a fossil fuel?
A) Wood B) Uranium C) Coal D) Water
17. Which gas causes maximum global warming?
A) Methane B) Carbon dioxide C) Nitrous oxide D) Water vapor

18. Which of the following ecosystems has the highest productivity?
A) Desert B) Tundra C) Tropical rainforest D) Grassland
19. Which law deals with water pollution control?
A) Water Act 1974 B) Air Act 1981 C) EPA 1986 D) Forest Act 1927
20. Noise pollution limit in residential areas during night is:
A) 75 dB B) 65 dB C) 70 dB D) 45 dB
21. Which one is not a fossil fuel?
A) Oil B) Coal C) Natural gas D) Solar energy
22. Human activities have the most impact on which sphere?
A) Atmosphere B) Biosphere C) Lithosphere D) Hydrosphere
23. Which of the following is a keystone species?
A) Rabbit B) Tiger C) Deer D) Peacock
24. Which gas is produced in landfills?
A) CO₂ B) Methane C) Ozone D) Oxygen
25. EIA stands for:
A) Environmental Impact Assessment B) Energy Information Analysis
C) Environmental Improvement Act D) Environmental Impact Act
26. Hydroelectric power uses:
A) Sunlight B) Wind C) Water D) Biomass
27. In India, Forest Survey is done every:
A) 2 years B) 2 years C) 5 years D) Annually
28. "Green Revolution" is related to:
A) Forest growth B) Industrial development
C) Agricultural productivity D) Environmental conservation

Group-B

(Answer any 6 questions from the following)

(Each question carries 10 marks)

6 × 10 = 60)

1. What is an ecosystem? Discuss its structure and functions?
2. What are ecological pyramids? Discuss types of ecological pyramids with examples.
3. Explain causes, impacts, and control measures of water pollution.
4. Discuss the causes of biodiversity loss and strategies for biodiversity conservation.
5. Define and distinguish between renewable and non-renewable resources with examples.
6. Write a detailed note on the Environmental Protection Act, 1986.
7. Discuss global warming and measures to control it.
8. What is sustainable development? Explain its objectives and importance.
9. Explain acid rain: causes, effects, and preventive measures.
10. Write short notes on:
 - a) Ecological succession
 - b) Phosphorus cycle

ALIAH UNIVERSITY

Regular End Semester Examination (Spring Semester)2025 (B. Tech 2nd Year 4th Semester)

Subject Name: Object Oriented Programming Systems

Full Marks: 80

Time: 3 Hours

Subject Code: CSEUGPC05

Group A

Answer all 5 questions

1. Answer the following briefly 5 X 2= 10
- What do you mean by applet in Java? (CO6- Understand)
 - What do you mean by data abstraction? (CO1- Understand)
 - How do you throw an exception in Java? (CO6- Understand)
 - What do you mean by an Interface? (CO5- Understand)
 - What is the difference between class and abstract class in Java? (CO4- Analyse)

Group- B

Answer any 6 questions

6X 5 = 30

- How to create your own package in Java? (CO5- Understand)
- Draw a table to explain the concept of Access Modifiers in Java. (CO5- Remember)
- Describe a class variable in Java with suitable example. (CO2- Understand)
- Explain the different types of compile time and run time errors. (CO6- Remember)
- What is a constructor? Explain the concept of constructor overloading with an example. (CO3- Understand) 2+3
- Write a Java Program with output to explain what will happen if we call the run () method directly without calling the start () method in a thread. (CO4- Analyse)
- Explain the concept of inheritance. Why is it so useful in Java? (CO4- Understand) 2+3

Group- C

Answer any 4 questions

4X 10 =40

- Explain the concept of Exception Handling in Java with a suitable example. Define Applet with an Example. 5+5 (CO6- Understand)
- Write a program in Java to create two threads such that the first thread will print all Fibonacci numbers in a range and the second thread will print all the cubes of natural numbers in a range. 5+5 (CO5- Apply)
- Explain the concepts of Overloading, Overriding and Dynamic Method Dispatch in Java. 3+3+4 (CO4- Understand)
- With a help of a neat state transition diagram explain the life cycle of a thread in Java. (CO4- Understand)
- What do you mean by "super"? Discuss any seven features of Object-Oriented Programming. 3+7 (CO4- Understand) (CO1- Understand)

ALIAH UNIVERSITY
End Semester Examination (Spring Semester) 2024
(B.Tech CSE 2nd Year 4th Semester)

Subject Name: Object Oriented Programming Lab

Full Marks: 100

Time: 3 Hours

Subject Code: CSEUGPC07

Answer any one question

1. Write Java Program to print all the Prime numbers and Armstrong Numbers in a range given by the user. Take input through BufferedReader class object.
2. Write a Java Program to create a class called Complex. Within the class define the relevant data variables as well as methods to sum, subtract multiply and divide two complex numbers. Create Complex class objects in main methods and perform summation and multiplication.
3. Write a Java Program to take input of seven city name through command line argument. Sort these cities using selection sort in alphabetic order by using compareTo method.
4. Write a Java Program to take a string input by using BufferedReader class object. Reverse this string and print it.
5. Define an Abstract class Shape and its two Subclasses Square and Circle. Use method overriding to calculate the area of the square and circle.
6. Define a class Shape and within the class define overloaded methods calculateArea to calculate the area of Circle and Square. Use BufferedReader for input.
7. Define an interface Shape with abstract method calculatePerimeter. Implement the interface in two classes called Square and Circle. Create its objects to calculate perimeter and area.
8. Use multithreading to create two threads Fibonacci and Prime. Fibonacci thread will display the series for the n terms with 2000ms delay and Prime thread will display prime numbers of a given range with 5000ms delay.
9. Write a Java program to handle "divide by zero exception" with the following message- "Exception Occurred" if the quotient is -1. Otherwise print the result in finally block.
10. Define a class Students with marks as one of the data members. In the main method create an array of Student Object and calculate the obtained average marks by all the students. Take input using Scanner class object.